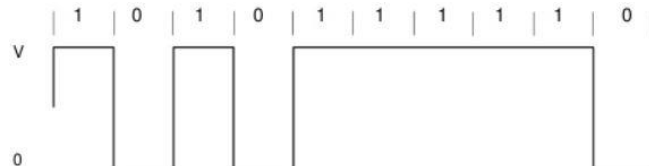


Department of Information and Communication Engineering
ICE-3206 (Digital Communication Sessional)
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1. Unipolar NRZ Signaling:

In this line code, symbol 1 is represented by transmitting a pulse of amplitude A /voltage V for the duration of the symbol, and symbol 0 is represented by switching off the pulse, as illustrated in the following figure:



Matlab Code:

%Unipolar Non Return to Zero Line Coding

clc;

clear all;

close all;

N=10;

%Number of bits

n=randi([0,1],1,N)

%Random bit generation

%Mapping Function

for m=1:N

if n(m)==1

nn(m)=1;

else

nn(m)=0;

end

end

nn

%Signal Shaping

i=1;

t=0:0.01:length(n);

%100 Times duration set up for a single binary bit

for j=1:length(t)

%Indexing set-up for time duration

if t(j)<=i

%Binary input data index Check-up Condition

y(j)=nn(i);

%Assign value from the mapping function

else

y(j)=nn(i);

i=i+1;

%Binary input data index increment

end

end

plot(t,y, 'linewidth',2);

axis([0,N,-1.5,1.5]);

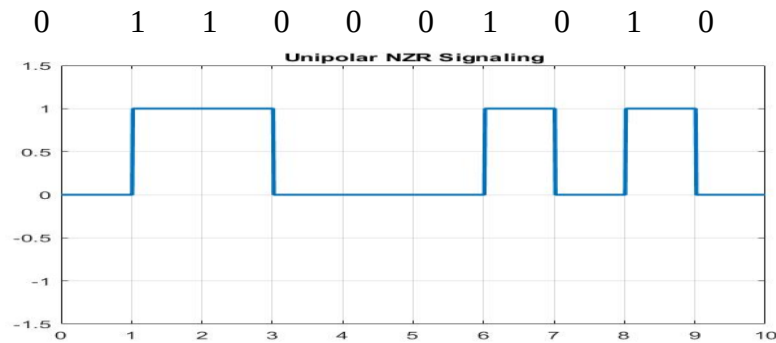
%Axis set-up

grid on;

title("Unipolar NZR Signaling");

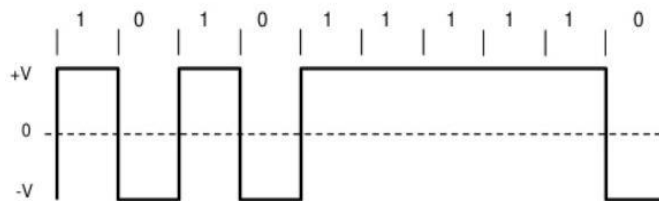
Output:

Binary Data: n =



2. Polar NRZ Signaling:

In this line code, symbols 1 and 0 are represented by transmitting pulses of amplitudes $+A$ and $-A$, respectively, as illustrated in the following figure:



Matlab Code:

%Polar Non Return to Zero Line Coding

clc;

clear all;

close all;

N=10;

%Number of bits

n=randi([0,1],1,N)

%Random bit generation

%Mapping Function

for m=1:N

if n(m)==1

nn(m)=1;

else

nn(m)=-1;

end

end

nn

%Signal Shaping

i=1;

t=0:0.01:length(n);

%100 Times duration set up for a single binary bit

for j=1:length(t)

%Indexing set-up for time duration

if t(j)<=i

%Binary input data index Check-up Condition

y(j)=nn(i);

%Assign value from the mapping function

else

y(j)=nn(i);

i=i+1;

%Binary input data index increment

end

end

plot(t,y, 'linewidth',2);

axis([0,N,-1.5,1.5]);

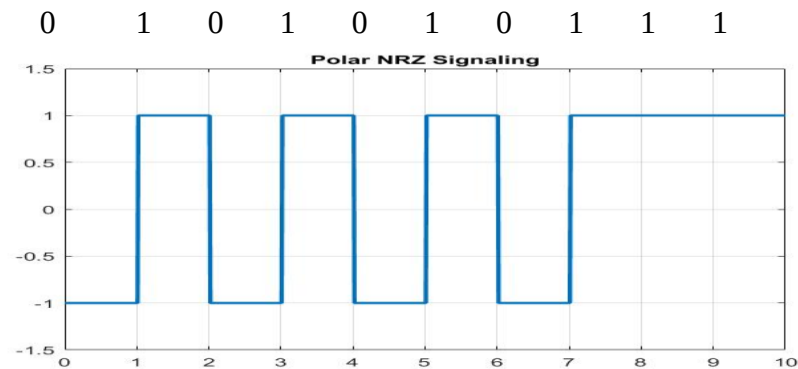
%Axis set-up

grid on;

title("Polar NZR Signaling");

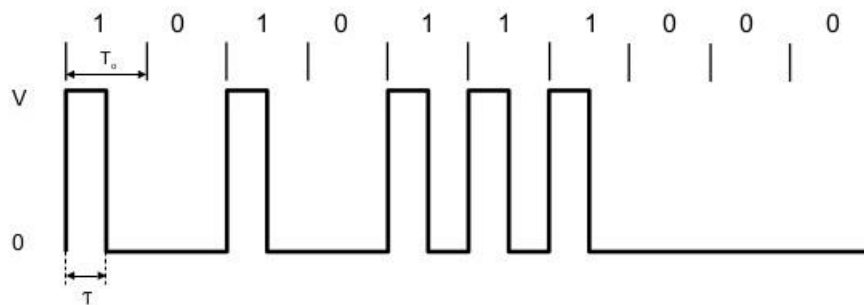
Output:

Binary Data: n =



3. Unipolar RZ Signaling:

In this line code, symbol 1 is represented by a rectangular pulse of amplitude A / voltage v for half-symbol width, and symbol 0 is represented by transmitting no pulse, as illustrated in figure:



Matlab Code:

%RZ Unipolar line coding

clc;

clear all;

close all;

N=10;

n=randi([0,1],1,N)

%RZ Pulse Shaping

i=1;

a=0;

b=0.5;

t=0:0.01:length(n);

for j=1:length(t)

if t(j)>=a && t(j)<=b

y(j)=n(i);

elseif t(j)>b && t(j)<=i

y(j)=0;

else

i=i+1;

a=a+1;

b=b+1;

end

end

plot(t,y,'lineWidth', 2);

axis([0,N,-1.5,1.5]);

grid on;

title('Unipolar return-to-zero (RZ) signaling');

%Number of bits

%Random bit generation

%Initial value for the first half cycle

%Initial value for the second half cycle

%Condition for the first half cycle

%Assign first 50 values for

%Condition for the Second half cycle

%Set all values 0 for the second half cycle

%Binary input data index increment

%Initial value for the first half cycle increment

%Initial value for the second half cycle increment

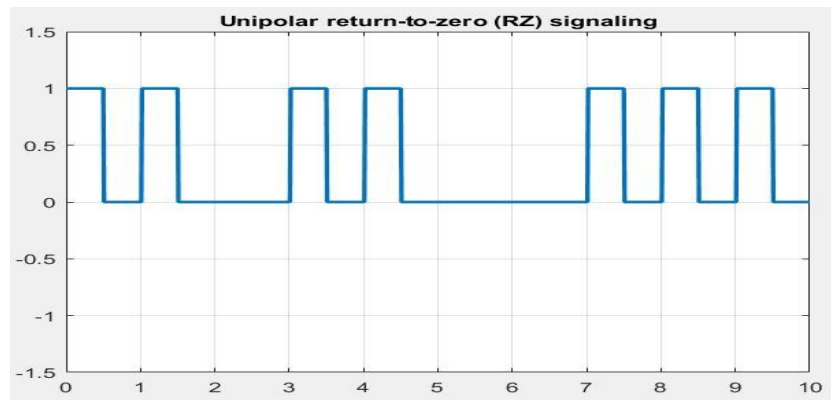
%Linewidth 2 for clear visualization

%Axis set-up

Output:

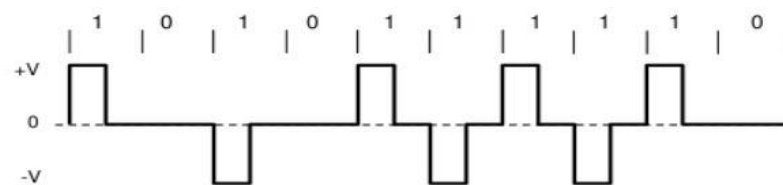
Binary Data: n =

1 1 0 1 1 0 0 1 1 1



4. Bipolar RZ Signaling:

This line code uses three amplitude levels as indicated in the following figure. Specifically, positive and negative pulses of equal amplitude (i.e., $+A$ and $-A$) are used alternatively for symbol 1, with each pulse having a half-symbol width; no pulse is always used for symbol 0.



Matlab Code:

%Bipolar Return to Zero Signaling

clc;

clear all;

close all;

N=10; %Number of bits

n=randi([0,1],1,N) %Random bit generation

%Binary to Bipolar Conversion

f=1;

for m=1:N

if n(m)==1

if f==1

nn(m)=1;

f=-1;

else

nn(m)=-1;

f=1;

end

else

nn(m)=0;

end

end

nn

%Bipolar RZ Pulse Shaping

i=1;

a=0; %Initial value for the first half cycle

b=0.5; %Initial value for the second half cycle

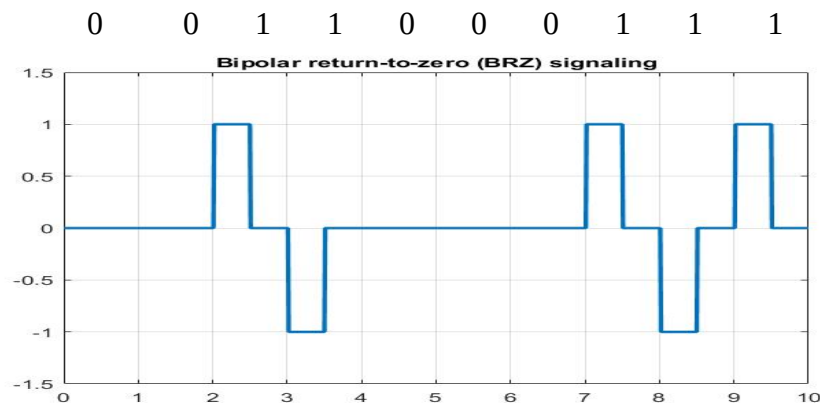
```

t=0:0.01:length(n);
for j=1:length(t)
if t(j)>=a && t(j)<=b    %Condition for the first half cycle
    y(j)=nn(i);          %Assign first 50 values for
elseif t(j)>b && t(j)<=i %Condition for the Second half cycle
    y(j)=0;              %Set all values 0 for the second half cycle
else
    i=i+1;               %Binary input data index increment
    a=a+1;               %Initial value for the first half cycle increment
    b=b+1;               %Initial value for the second half cycle increment
end
end
plot(t,y,'lineWidth', 2); %Linewidth 2 for clear visualization
axis([0,N,-1.5,1.5]);    %Axis set-up
grid on;
title('Bipolar return-to-zero (BRZ) signaling');

```

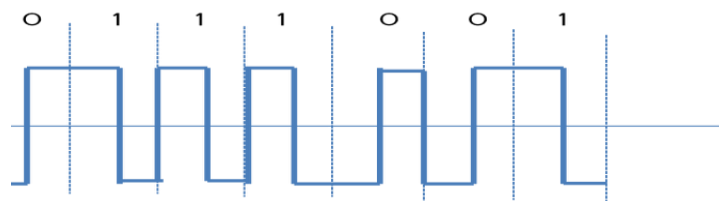
Output:

Binary Data: n =



5. **Split-Phase (Manchester Code):**

In this method of signaling, illustrated in the figure, symbol 1 is represented by a positive pulse of amplitude A followed by a negative pulse of amplitude $-A$, with both pulses being half-symbol wide. For symbol 0, the polarities of these two pulses are reserved.



Matlab Code:

```

%Split Phase-Manchester Coding
clc;
clear all;
close all;
N=10; %Number of bits
n=randi([0,1],1,N) %Random bit generation
%Binary to Manchester Conversion
nnn=[];
for m=1:N
    if n(m)==1
        nn=[1 -1];
    else
        nn=[-1 1];
    end
    nnn=[nnn; nn];
end

```

```

else
    nn=[-1 1];
end
nnn=[nnn nn];
end
nnn
%Manchester Coding Pulse Shaping
i=1;
l=0.5;
t=0:0.01:length(n);
for j=1:length(t)
    if t(j)<=l
        y(j)=nnn(i);
    else
        y(j)=nnn(i);
        i=i+1;
        l=l+0.5;
    end
end
plot(t,y,'LineWidth', 2);
axis([0,N,-1.5,1.5]);
grid on;
title('Manchester Coding');

```

%Condition for the first half cycle
%Assign first 50 values for
%Linewidth 2 for clear visualization
%Axis set-up

Output:

Binary Data: n =

